

Pratham Vishwakarma

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Professional Summary

Software Development Engineer I with production experience at Jio Platforms Limited, specializing in Python, FastAPI, and AI-driven infrastructure automation (MCP). Enabled LLM agents to execute 50+ production operations autonomously, reduced manual effort by 50% across 3 teams, and accelerated automation throughput by 60%. Leveraging strong DevOps and Kubernetes expertise from independent projects, targeting DevOps, Backend Engineering, or Cloud Infrastructure roles.

Technical Skills

Languages & Backend: Python, FastAPI, Shell Scripting

DevOps & Cloud: Docker, Kubernetes, Jenkins, ArgoCD (GitOps), Terraform, AWS, GCP

Observability & Tools: Prometheus, Grafana, SaltStack, Linux, Git, GitHub, Azure Repos

Databases: PostgreSQL, MySQL

AI Infrastructure: Model Context Protocol (MCP)

Professional Experience

Software Development Engineer I | Jio Platforms Limited

Navi Mumbai, MH | Dec 2023 - Present

Product Engineering - Jio CloudXP / Hybrid Cloud Management Platform

- **AI Infrastructure Orchestration (MCP):** Built a FastAPI MCP server exposing 50+ tools, enabling AI agents to autonomously execute production operations (reducing MTTD by 40%). Secured LLM actions via Piko reverse tunnels and a custom AuthZ layer.
- **SaltStack Statefile Management:** Developed and maintained state files to enforce strict configuration compliance across hundreds of servers, eliminating drift.
- **Automation Development & Reusable Workflow Packs:** Developed modular automation workflow packs using Python and Robot Framework for Jio CloudXP, increasing automation delivery speed by 60% and reducing manual execution by 50%.
- **Backend Development & Cross-Platform Execution Logic:** Engineered cross-platform backend execution logic for the RPA module across Linux (SSH/SCP) and Windows (WinRM), ensuring 100% compatibility and reducing development effort by 25%.

Key Achievements

- **AI Infrastructure Orchestration (MCP):** Built a FastAPI MCP server exposing 50+ tools, enabling AI agents to execute production operations (reducing MTTD by 40%). Secured actions via Piko tunnels and a custom AuthZ layer.
- **SaltStack Statefile Management:** Developed and maintained SaltStack state files to enforce strict configuration compliance across hundreds of servers, eliminating drift.
- **Automation Development & Reusable Workflow Packs:** Created modular automation workflow packs with Python & Robot Framework, boosting delivery speed by 60% and cutting manual effort by 50%.
- **Backend Development & Cross-Platform Execution Logic:** Engineered cross-platform backend for RPA using Linux SSH/SCP and Windows WinRM, achieving 100% compatibility and reducing development effort by 25%.

Projects

Notes-Studio - Cloud-Native Microservices Application

March 2026 - Present

www.prathamvishwakarma.com/notes

- **Architecture and Kubernetes:** Engineered a containerized 3-tier microservices application using FastAPI, PostgreSQL, and Next.js with JWT-based authentication. Deployed on Kubernetes using Persistent Volume Claims (PVCs) and Secrets for secure state management.
- **CI/CD and Observability:** Built zero-downtime Jenkins pipelines with automated rollback and integrated Prometheus and Grafana for real-time API performance and infrastructure monitoring across DEV, SIT, and PROD environments.

Yashvi Duniya - Live News Platform

Nov 2025 - Jan 2026

www.yashasviduniya.com

- **DevOps and Infrastructure:** Architected cloud infrastructure and automated CI/CD pipelines for a high-traffic live news portal, implementing system monitoring and caching strategies to ensure maximum uptime during news traffic spikes.

Education

Bachelor of Technology (B.Tech) in Computer Science Engineering

2019 - 2023

